Joint Role–View Specialization

Frozen concept grounding · role-specific evidence use · aligned score-field fusion

(a) Role-functional text

Official anchor
“road” + official aliases

Presence
“Road network”

Semantic
“continuous paved road surface”

Instance
“connected road region”

Fixed dataset profile π: selected role slots and view operator; shared by both observations

(b) Observations



Local observation



Context observation

(c) Grounding and composition

Frozen SAM3
Each required prompt → full P / S / I outputs

Role updates
Bounded updates + native p × max(S, I)

Restore
Official map + coordinates

Frozen SAM3
Each required prompt → full P / S / I outputs

Role updates
Bounded updates + native p × max(S, I)

Restore
Official map + coordinates

(d) View fusion

GL

GG

Ψ_ω

Decision
Argmax + background rule

Segmentation $\hat{Y}$

Same weights and role profile in both rows • Fusion uses complete class-score fields before argmax

Anchor-preserving role composition

Apply this same computation to Local and Context; the SAM3 weights remain frozen.

(a) Complete grounding

Full frozen SAM3 per prompt

Shared weights; separate grounding runs

Prompt	P	S	Queries
Anchor A	p^A	S^A	Q^A
Presence πP	p^P	S^P	Q^P
Semantic πS	p^s	S^s	Q^s
Instance πI	p^I	S^I	Q^I

Colored cells indicate evidence used downstream.

(b) Selective evidence updates

Presence update

$\tilde{p} = B(p^A, p^P)$

Semantic update

$\tilde{S} = B(S^A, S^s)$

Presence controls instance construction

Instance reconstruction + update

Rebuild anchor / candidate maps using $\tilde{p}$

$\tilde{I} = B(I^A(\tilde{p}, a), I^I(\tilde{p}, a))$

(c) Native composition

Native fusion

$F\pi = \tilde{p} \times \max(\tilde{S}, \tilde{I})$

Official-map correction

$G_{v,\pi} = \mathrm{clip}(G^o_v + F_{v,\pi} - F^A_v)$

Preserves official aliases
and baseline aggregation order

$$B(a, b) = \mathrm{clip}_{[0,1]}\,[a + \alpha \cdot \mathrm{clip}(b - a, -\tau, \tau)] \qquad \alpha = 0.5,\ \tau = 0.25$$

Admission $a = \tilde{p}$ (native) or p^A (anchor admission). Zero role updates recover the official map for that observation.